

FROM GROUND FRAGMENT DATA TO BOTTOM UP DECISION MAKING

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Many situations of a different nature require to be qualified globally with respect to a single criterion with a yes or no answer in order to decide on an eventual action. It goes from the evaluation of a new bank saving product to the existence of weapons of mass destruction in a foreign country. In each case there is no direct access to the global information spared on the ground. It has to be constructed from a series of successive collecting and synthesizing local information, which is scattered, fragment, fuzzy and incomplete. A puzzle has thus to be reconstituted from the ground level up to the institution top command decision level, always rather far from the ground.

The instrumental question is to determine if such a procedure is reliable and if it obeys to some universal mechanism. The current problem of Iran nuclear activities illustrates the question for instance with American intelligence agency claiming military program has been stopped after 2003 and others agencies claiming the opposite. The same occurred with Iraq before the American invasion with US and British Intelligences claiming there were WMD against France and Russia Intelligences claims. The central issue being the eventual discrepancy between the reality and its bottom up construction.

It is of importance to determine if such decisions are based on wrong conclusions from Intelligence services or if they were motivated by political reasons driven lies. In particular knowing that both sides used the same information and reports to draw entirely different conclusions. We adopt the point of view that neither side was necessarily lying.

Using a toy model we provide a rational explanation to such a paradox of getting opposite conclusions from the same "rough ground data". It is shown that any bottom-up collecting Information gets naturally biased from the very nature of the process driven by the necessary motivation behind any investigation which is always conducted from an a priori. On this basis it is found that any systematic principle of precaution or risk taken may turn very dangerous in evaluating a global risk. Depending on the degree of danger on the ground, it is found that there exists two regimes, one where the final picture is right and one where it is wrong. These results enlighten the possible catastrophic effects of a top decision for an action decided upon the bottom up data signature of the ground situation.

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